

1. Research Questions

- Q1 (Proportion): Is there a significant difference in the proportion of current alcohol use between male and female high school students?
- Q2 (Mean): Among students who drink, does the mean frequency of alcohol use differ by gender?

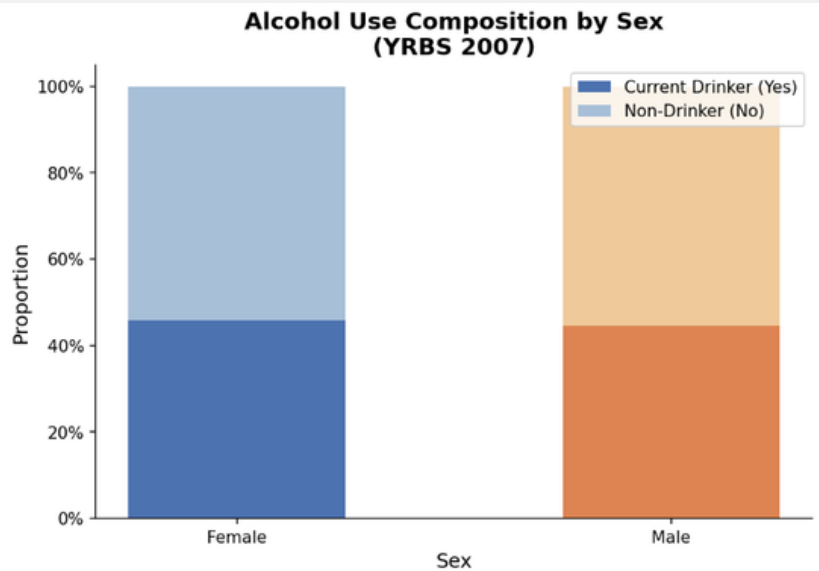
2. Data & Variables

Dataset: YRBS 2007 after listwise deletion N = 12,659

- Grouping Variable: **WhatIsYourSex** (Male: 6,425 & Female: 6,234)
- Response Variable: **CurrentAlcoholUse** (Original: 1-7 ordinal scale).

3. EDA

Male and female students show a visually similar initial baseline in overall drinking status, but distinct distributions emerge when plotting the exact 1–7 frequency levels.

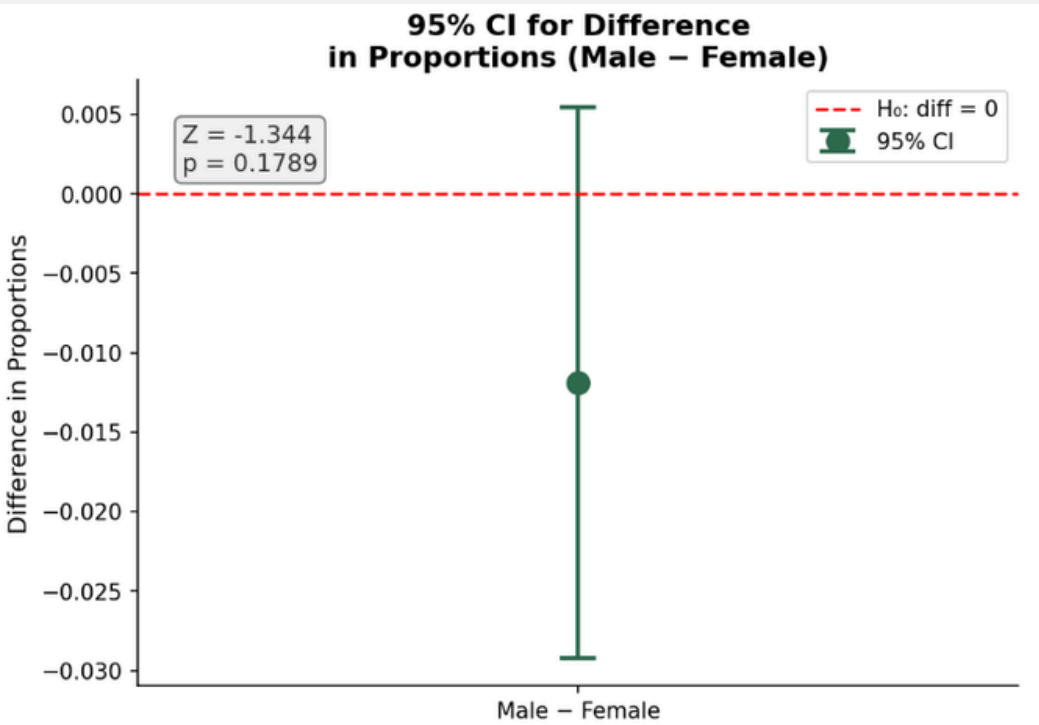


4. Method Selection

- For Proportion : **Two-Proportion Z-Test** (CurrentAlcoholUse recoded to Binary: 0 = 0 days, 1 = ≥ 1 day).
- For Mean : **Welch's Two-Sample T-Test** (On original 1–7 scale for current drinkers to account for unequal variances).

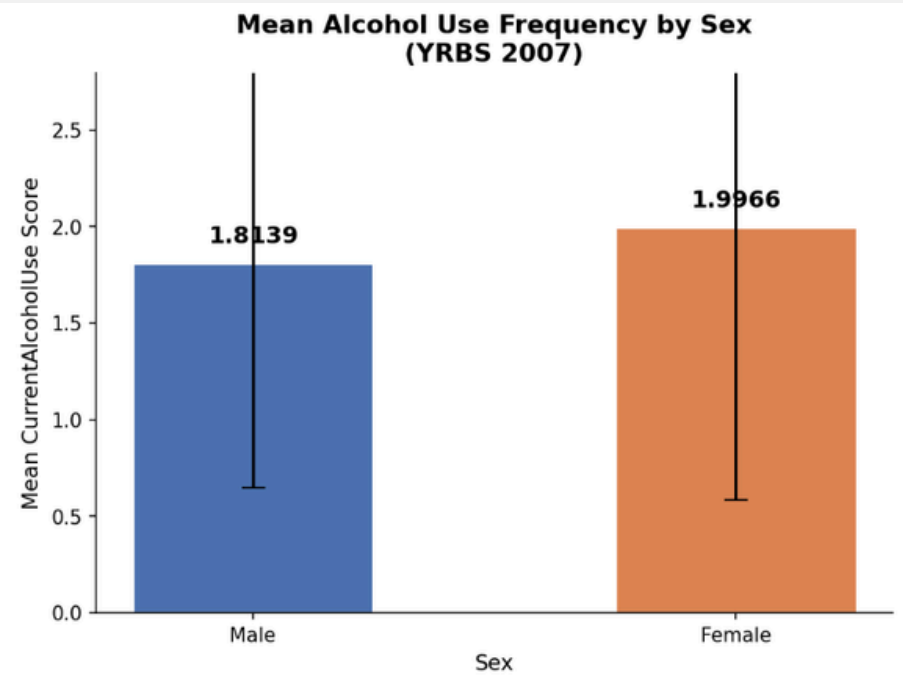
5. Key Results: Proportion Analysis

- Sample Estimates:
Male $\hat{p}_1 = 52.73\%$
Female $\hat{p}_2 = 50.12\%$
Difference $(\hat{p}_1 - \hat{p}_2) = +1.19\%$
- Test Result: $Z = -1.3442$, $p\text{-value} = 0.1789$
- 95% Confidence Interval (CI): $(-0.0292, 0.0054)$



6. Key Results: Mean Frequency Analysis

- Sample Estimates:
Male Mean $(\bar{x}_1) = 1.8139$ (SD = 1.34)
Female Mean $(\bar{x}_2) = 1.9966$ (SD = 1.38)
Difference $(\bar{x}_1 - \bar{x}_2) = -0.1827$
- Test Result: $T = -7.9389$, $p\text{-value} < 0.0001$
- 95% Confidence Interval (CI): $(-0.2278, -0.1376)$



7. Final Conclusion

Statistical Decisions:

1. Proportion: Fail to Reject H_0 ($\alpha=0.05$)
2. Mean Frequency: Reject H_0 ($\alpha=0.05$)

The Meaning:

"Gender doesn't tell us if a student drinks, because the overall percentage is the same. However, among regular drinkers, females drink more often than males."